

MUJTABA TARIQ

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Summary

AI and Full-Stack Developer with experience in machine learning, LLM fine-tuning, automation systems, web scraping, and end-to-end application development. Skilled in building AI-powered tools using Python, React.js, Node.js, Flask, APIs, and ML frameworks. Experienced in computer vision data annotation, workflow automation, model integration, and deploying practical software solutions with Docker, REST APIs, and database-backed architectures.

Skills

Languages: Python, JavaScript, C++, SQL

AI / Machine Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, Transformers, PEFT, QLoRA, XGBoost, KNN, K-means, CNNs, Model Fine-Tuning

Web Development: React.js, Node.js, Express.js, Flask, Streamlit, REST APIs, JWT, HTML, CSS

Databases: PostgreSQL, MongoDB, MySQL, Prisma ORM

Automation / Scraping: Playwright, Google Apps Script, API Integrations, Web Scraping, Workflow Automation

Tools: Git, GitHub, Docker, Postman, Visual Studio Code, Cloudinary, Google Maps API, Meta Graph API

Experience

Data Annotator Jan 2026 – Present
Motive Islamabad, Pakistan

- Annotate large-scale video datasets using bounding boxes for object detection, scene understanding, and motion tracking tasks.
- Perform frame-level quality checks to improve annotation consistency across dynamic visual scenes.
- Follow structured labeling guidelines to support computer vision model training and dataset reliability.
- Collaborate within a pod-based workflow to maintain annotation throughput, task accuracy, and SLA compliance.
- Designed an internal workflow optimization process to improve task allocation efficiency and overall pod performance.

AI Solutions Developer Jun 2025 – Sep 2025
Rebiz (Oxiliry) Remote / Pakistan

- Designed AI-driven automation systems for daily summaries, email organization, operational monitoring, and weekly reporting.
- Integrated external APIs including news feeds, uptime monitors, search sources, and Shopify Admin API into automated business workflows.
- Built a multi-source data pipeline combining Shopify data, search results, and local LLM inference for structured reporting.
- Developed competitor price matching, data normalization, and reporting logic for pricing intelligence use cases.
- Containerized automation workflows with Docker and exposed backend services through Flask REST APIs.
- Delivered dashboards and structured reports for compliance, pricing, and operational decision-making.

MERN Stack Developer Mar 2025 – May 2025
Macro Soar Pakistan

- Worked on ormo.ai, an AI-powered consultancy platform for consultant registration and custom AI chatbot deployment.
- Developed React.js interfaces for onboarding, consultant registration, chatbot setup, and user-facing workflows.
- Built and integrated Node.js and MongoDB backend services to manage users, consultancy sessions, and application data.
- Supported the organization's website, database operations, and day-to-day technical workflows.

Front-End Developer Intern Sep 2024 – Feb 2025
Digital Empowerment Pakistan

- Developed a React-based search engine using an external API with structured state management and reusable components.
- Used React hooks and Context API to manage global state, optimize data fetching, and improve application maintainability.
- Implemented dynamic form handling, controlled components, and client-side validation for user input workflows.

JavaScript Developer Jul 2024 – Sep 2024
Insafdaar Pakistan

- Led a small development team to design and deliver Google Workspace automation solutions.
- Developed Google Apps Script automations for Drive file management, Google Sheets workflows, Gmail alerts, and Calendar scheduling.

- Integrated third-party APIs with Google Apps Script to extend functionality across internal tools.
- Built dynamic web forms linked with Google Sheets for real-time data collection, tracking, and reporting.
- Automated repetitive administrative workflows, reducing manual effort and improving data consistency.

Projects

Medical AI Assistant Bot

AI Assistant / Web Scraping / Healthcare Automation

- Built an AI-powered medical assistant that collects user symptoms, asks severity-based follow-up questions, and guides users toward relevant medical support.
- Implemented a scraping workflow to collect top doctor data from Marham and use it for doctor discovery and recommendation flows.
- Designed a structured triage-style conversation flow to separate general guidance, urgent symptoms, and doctor recommendation scenarios.
- Developed the system as a practical healthcare assistant focused on symptom understanding, doctor matching, and user-friendly interaction.
- Key technologies: Python, Web Scraping, LLM Integration, REST APIs, Healthcare Automation.

Smart Construction Model

Full-Stack ML System

[GitHub Repository](#)

- Developed a web-based platform for construction cost estimation, real estate recommendation, and society analysis.
- Built a cost estimation module to calculate construction expenses using area, material category, labor, and cost-tier inputs.
- Designed a society search and recommendation system using Google Maps, geolocation data, K-means clustering, and KNN.
- Integrated XGBoost-based prediction models to estimate house and plot price trends from historical data.
- Implemented secure authentication with JWT, an admin panel, and a PostgreSQL database using Prisma ORM.
- Key technologies: React.js, Node.js, Flask, PostgreSQL, Prisma, XGBoost, K-means, KNN, Google Maps API, JWT.

Phi-3 Mini Marketing AI Assistant

LLM Fine-Tuning / QLoRA

- Fine-tuned Phi-3 Mini 4k-Instruct into a domain-specific marketing assistant for product captions, ad copy, slogans, and brand-focused content.
- Created a curated dataset of approximately 400 structured marketing samples using system, user, and assistant message formats.
- Applied 4-bit NF4 quantization, QLoRA, PEFT, gradient checkpointing, and paged AdamW to fine-tune the model locally on a GTX 1650 4GB GPU.
- Saved reusable LoRA adapter weights for future inference, merging, or additional fine-tuning.
- Tested base vs fine-tuned outputs and observed improved tone control, marketing style, and brand-specific phrasing.
- Key technologies: Python, Hugging Face Transformers, PEFT, QLoRA, Phi-3 Mini, 4-bit Quantization, CUDA.

Protein Antigenicity Prediction System

Bioinformatics / Deep Learning

[GitHub Repository](#)

- Developed a Flask-based web application for predicting protein antigenicity from biological sequence input.
- Implemented FASTA sequence handling and protein feature extraction for immunogenicity prediction workflows.
- Used UniRep-style protein embeddings and a CNN-based model architecture to support sequence classification.
- Displayed prediction confidence and model results through a responsive web interface.
- Prepared the application for deployment using Docker, Gunicorn, and a structured Python project setup.
- Key technologies: Python, Flask, PyTorch, TensorFlow, CNN, UniRep Embeddings, FASTA, Docker.

X-ray Image Classification

Computer Vision / Machine Learning

- Built a web application for classifying X-ray images as Normal or COVID-19 using a machine learning pipeline.
- Preprocessed X-ray images with OpenCV, including RGB-to-grayscale conversion, resizing, normalization, and feature preparation.
- Trained a KNN classifier and integrated the trained model into a Flask backend for real-time image classification.
- Displayed model evaluation outputs including accuracy, confusion matrix, and ROC curve through the application interface.
- Key technologies: Python, Flask, OpenCV, Scikit-learn, KNN, Computer Vision.

Prayer Time Calculation Engine

CLI / Astronomical Calculations

- Developed a command-line prayer time engine to calculate Fajr, Dhuhr, Asr, Maghrib, and Isha timings from location-based inputs.
- Implemented astronomical calculations including solar declination, equation of time, solar noon, hour angles, and angle-based prayer rules.
- Designed a lightweight CLI workflow for quick local use without depending on external scraping services.
- Key technologies: Node.js / Python, CLI Development, Astronomical Algorithms, Date-Time Calculations.

Additional GitHub Projects

Automation / API Integration / Scraping

- [Solace Pixel](#): Built a Streamlit tool for generating portrait social media posts, uploading images to Cloundinary, and publishing to Instagram/Facebook through Meta Graph API.

- [Terminal-Based Daraz Scraper](#): Created a Playwright-based CLI scraper to extract product titles, prices, ratings, reviews, screenshots, and JSON outputs for market research and price comparison.

Education

Bahria University Mar 2021 – Feb 2025
Bachelor of Science in Information Technology Islamabad, Pakistan

- CGPA: 3.25 / 4.00
- Relevant coursework: Information Security, Cybersecurity, Software Testing, Software Requirements Engineering, Networking, Quality Assurance, Object-Oriented Programming, Data Structures, Database Design and Management.

Pakistan School 2018 – 2020
Higher Secondary Certificate (FBISE) Isa Town, Bahrain

- Grade: B
- Subjects: English, Mathematics, Physics, Computer Science.